Intelli■Credit: AI■Powered Corporate Credit Analysis (Hackathon Prototype)

1. Problem Statement

Banks evaluate corporate borrowers by reviewing financial statements, GST filings, bank statements, annual reports, and external information such as news and litigation records. Credit officers manually compile this information into a Credit Appraisal Memo (CAM) to decide whether to approve a loan. This process is slow, fragmented, and prone to human oversight, especially when important signals are hidden inside long financial documents.

2. Project Objective

The goal of Intelli■Credit is to demonstrate how Artificial Intelligence can assist credit officers by automatically analyzing company documents, extracting financial insights, detecting risk signals, and generating a structured Credit Appraisal Memo draft. This hackathon prototype focuses on automation of document analysis and risk summarization.

3. Our Solution

- Upload company financial documents such as annual reports or financial statements.
- Use AI and OCR to extract key financial values from documents.
- Calculate financial ratios and basic credit indicators.
- Collect external information such as company news and sentiment.
- Generate a simple risk score with explainable factors.
- Automatically create a draft Credit Appraisal Memo (CAM).

4. Hackathon System Architecture

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Company Document Upload
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Document Processing (OCR + PDF Parsing)
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        ▼
AI Extraction (LLM API)
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        ▼
Financial Ratio Analysis
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        ▼
External Research (News & Sentiment)
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        ▼
Risk Scoring Engine
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        ▼
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Explainable Risk Summary
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        ▼
CAM Report Generator
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5. Backend Workflow

- User uploads company financial documents through the application.
- Backend extracts text and tables using OCR and PDF parsing tools.
- An LLM API processes extracted text and identifies key financial figures such as revenue, profit, and debt.
- Financial ratios such as Debt■to■Equity and Profit Margin are calculated.
- External research APIs gather company news and perform sentiment analysis.
- A rule■based risk scoring engine calculates a credit risk score.
- The system generates an explainable summary of the main risk factors.
- A draft Credit Appraisal Memo (CAM) is generated automatically.

6. AI Models and Tools Used

- LLM APIs for document understanding and financial data extraction
- Tesseract OCR for scanned PDF text extraction
- Pandas and NumPy for financial calculations
- Sentiment analysis models for news interpretation
- Rule■based scoring logic for explainable credit risk estimation

7. Technology Stack

- Backend: FastAPI (Python)
- Document Processing: pdfplumber, pytesseract
- Data Processing: Pandas, NumPy
- AI Integration: LLM APIs
- Database: PostgreSQL or MongoDB
- Report Generation: python■docx / PDF generator

8. Expected Impact

This hackathon prototype demonstrates how AI can significantly reduce the time required to analyze corporate financial documents and prepare credit appraisal summaries. By automating document understanding and risk summarization, Intelli■Credit can help banks and financial institutions make faster and more informed lending decisions.

9. Future Improvements

- Improved financial data reconciliation across GST, bank, and financial statements

- Advanced machine learning models for probability■of■default prediction
- Integration with banking systems and loan origination platforms
- Post■sanction monitoring of borrower financial activity